Tutorial 1.2

Question 1: What is rectifier? How many types of rectifiers are there?

Question 2: What is half wave rectifier? Discuss the construction and principle of operation of a typical diode half wave rectifier.

Question 3: What is rectifier efficiency? Evaluate the efficiency of a typical half wave rectifier.

Question 4: What is a full wave rectifier? Discuss the construction and principle of operation of a typical centre tapped/bridge full wave rectifier.

Question 5: What is the ripple factor? Evaluate the ripple factor for both half-wave and full-wave rectifiers and discuss the performance.

Question 6: What is PIV? Why PIV is very important in diode circuits? Determine the PIV for the half-wave/full-wave/bridge rectifier circuit.

Question 7: What is a clipper/clamper circuit? How does a clipper/clamper circuit work? What are the applications of clipper/clamper circuits?

Question 8: What is a voltage doubler circuit? Illustrate the construction and principle of operation of a typical half-wave/full-wave voltage doubler circuit. What are their applications?

Problem 1:

EXAMPLE 2.18

(a) Sketch the output v_o and determine the dc level of the output for the network of Fig. 2.48.
(b) Repeat part (a) if the ideal diode is replaced by a silicon diode.
(c) Repeat parts (a) and (b) if V_m is increased to 200 V and compare solutions using Eqs. (2.7) and (2.8).

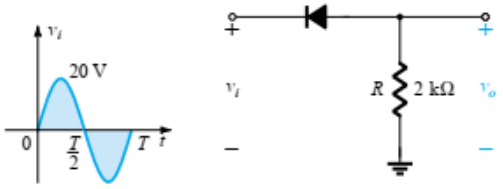


Figure 2.48 Network for Example 2.18.

Problem 2:

EXAMPLE 2.19

Determine the output waveform for the network of Fig. 2.63 and calculate the output dc level and the required PIV of each diode.

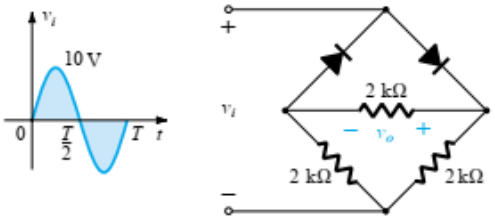


Figure 2.63 Bridge network for Example 2.19.

Problem 3:

EXAMPLE 2.20

Determine the output waveform for the network of Fig. 2.74.

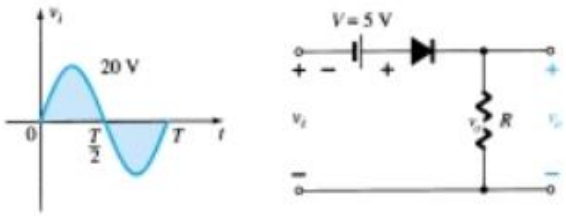


Figure 2.74 Series clipper for Example 2.20.

Problem 4:

EXAMPLE 2.2 Determine v_o for the network of Fig. 2.83.

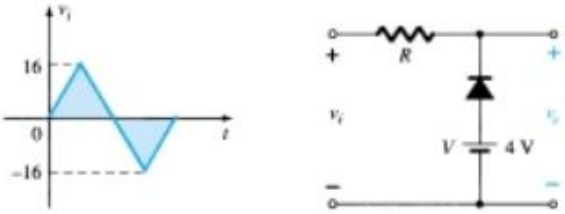


Figure 2.83 Example 2.22.